RAMRUP SATPATI

Data Science & Machine Learning Engineer | ENFJ-A Protagonist

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PROFESSIONAL SUMMARY

Degree-level BS student in Data Science & Applications at **IIT Madras** (Roll: 24f3004027 | **TOPPER - Top Course Grade - Diploma**). Technically grounded in Machine Learning, Deep Learning, Generative AI, and Linear Programming optimization. As an **ENFJ-A (Protagonist)**, blends mathematical and algorithmic rigor with empathetic collaboration, systems thinking, and clear communication to build scalable, human-centered computational applications.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) **2024 – Present**

Bachelor of Science (BS) in Data Science and Applications (Degree Level) **Chennai, India**

- **Academic Distinction:** Awarded **'TOPPER - Top Course Grade - Diploma'** badge; completed Foundational & dual Diploma coursework.
- **Key Coursework:** Machine Learning Foundations & Practice (MLP), Deep Learning & GenAI, Business Data Management (BDM), Modern App Dev I & II (MAD 1/2), Tools in Data Science (TDS).

TECHNICAL & LEADERSHIP SKILLS

- **Languages & Databases:** Python (Advanced), SQL (PostgreSQL, SQLite), JavaScript, Bash Scripting, HTML5/CSS3.
- **AI, ML & Modeling:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, GenAI & LLM architectures.
- **Optimization & Analytics:** Constrained Linear Programming (SciPy, Simplex), Customer Lifetime Value (CLV), Econometrics, EDA.
- **Frameworks & Tools:** Flask, Vue.js, Streamlit, RESTful APIs, Git, Docker, Linux/macOS, Jupyter.
- **Leadership Dynamics (ENFJ-A):** Empathetic leadership, psychological safety, milestone tracking, systems thinking.

FEATURED PROJECTS

Sales and Service Mix Optimization – Primary Data Analysis **Oct 2026**

Operations Research & Business Analytics | GitHub: 24f3004027/Business_Data_Management

- Executed primary field survey ($N = 220$) evaluating after-sales service (AMC) attachment economics across retail and enterprise buyers.
- Formulated and solved a constrained Simplex LP model balancing working capital and technician labor to optimize contribution margin.
- Identified optimal mix ($X^* = 74$ sales, $Y^* = 132$ AMCs), yielding **+28.4% profit uplift** (INR 1.226M/mo) and a 3.0x return on technician payroll.

Deep Learning & Generative AI Systems **2026**

Neural Architectures & Generative Modeling | GitHub: 24f3004027/Deep_Learning_Project

- Implemented deep learning pipelines utilizing CNNs, recurrent sequence models, and transformer architectures in PyTorch/TensorFlow.
- Applied transfer learning, latent space representations, and attention mechanisms to optimize inference latency and accuracy.

Placement Portal Application V2 (MAD 2 Capstone) **2025 – 2026**

Full-Stack Enterprise App | GitHub: 24f3004027/Placement_Portal_Application_V2

- Engineered an enterprise university placement platform with RBAC authentication for Admins, Corporate Recruiters, and Candidates.
- Integrated RESTful APIs with Flask, automated background job queues (Celery/Redis), and designed a reactive Vue.js interface.

Predictive Modeling & Statistical Machine Learning (MLP) **2025**

Supervised Learning & EDA | GitHub: 24f3004027/Machine_Learning_Practice_Project

- Developed end-to-end predictive classification pipelines in Scikit-Learn/XGBoost with grid-search hyperparameter tuning.
- Maintained monorepo utilities across multi-brand parking algorithms and hospital management systems (HMS).

EXPERIENCE & LEADERSHIP INITIATIVES

Open Source Contributor & Technical Project Lead **2024 – Present**

Independent Data Science & Software Developer | GitHub: RSNPIIT **Remote**

- Architected reproducible data science and optimization repositories; authored clear documentation and analytical codebooks.
- Applied ENFJ-A strengths to foster psychological safety, active team consensus, and structured milestone tracking in technical projects.